

# Convergence of the Discrete Consensus-Based Optimization Algorithm with Random Batch Interactions

Dongnam Ko

The Catholic University of Korea

Joint work with Seung-Yeal Ha, Shi Jin, and Doheon Kim

Shih-Hsien Fest on Hyperbolic Conservation Laws and Kinetic Theory,  
05 April, 2024

# Overview

- 1 Introduction; Consensus-Based Optimization
- 2 Random Batch Method (RBM) and its use on CBO
- 3 Previous works; kinetic equation for CBO
- 4 Main analysis; Contraction of diameter for random network
- 5 Summary

# Table of Contents

- 1** Introduction; Consensus-Based Optimization
- 2 Random Batch Method (RBM) and its use on CBO
- 3 Previous works; kinetic equation for CBO
- 4 Main analysis; Contraction of diameter for random network
- 5 Summary

# consensus-based optimization (CBO)

CBO: an evolutionary type of **gradient-free optimization algorithms** to find the minimum of a given cost function.

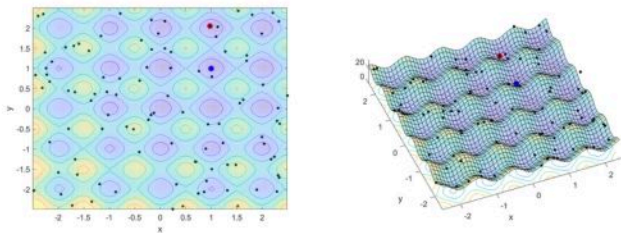
Similar methods; **Ant Colony Optimization, Particle Swarm Optimization, Wolf-pack optimization, Genetic algorithm, etc.:**

- 1** First, **spread the particles** into the domain.
- 2** Second, **evaluate current values** from particles' positions.
- 3** Third, process time-**evolution toward the possible minimum** positions.

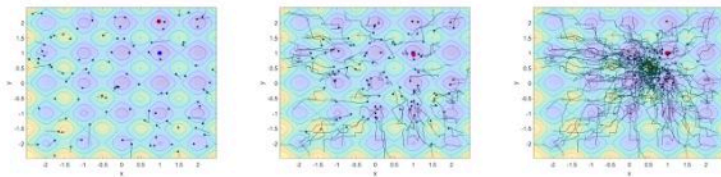
For a given  $L(x)$ , we want  $x_i(t)$  to approach  $x_* := \operatorname{argmin}_{x \in \mathbb{R}^d} L(x)$ .

# consensus-based optimization (CBO)

## SIMULATIONS:



**Figure:** Initial particle distribution and the Rastrigin cost function



**Figure:** Particle distribution at (left)  $t = 2$ , (middle)  $t = 10$ , (right)  $t = 50$

# consensus-based optimization (CBO)

Therefore, each particle **explores the domain**:

- $x_i = i$ -th agent's guess for  $\operatorname{argmin}_{x \in \mathbb{R}^d} L(x)$
- Iterate on  $t \in \mathbb{N}$ ,

$$x_i(t+1) = x_i(t) + (\text{interaction with other } x_j(t)\text{'s}), \quad i = 1, \dots, N$$

Three questions:

- 1 **[Consensus]**  $x_i(t) - x_j(t)$  decays to zero.
- 2 **[Convergence]** all  $x_i(t)$  converge to its limit  $x_i(\infty)$ .
- 3 **[Optimality]**  $x_i(\infty) \approx \operatorname{argmin}_{x \in \mathbb{R}^d} L(x)$  for some  $i$ .

# consensus-based optimization (CBO)

CBO is a kind of simplified version of multi-point optimization methods.

Algorithm [K.–Ha–Jin–Kim 2022] based on [Carrillo–Jin–Li–Zhu 2021]

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

$$\gamma > 0, \quad \eta_t^{i,\ell} \sim \mathcal{N}(0, \sqrt{\zeta}) \quad \text{for each } i, \ell, t \quad \text{and}$$

$$\bar{X}_t^{i,*} := \operatorname{argmin}_{x \in \{X_t^j | j \in N_i(t)\}} L(x), \quad N_i(t) \subset \{1, 2, \dots, N\}.$$

This time-evolution corresponds to a continuous stochastic dynamics:

$$dX_t^i = \lambda(\bar{X}_t^{i,*} - X_t^i)dt + \sigma \text{diag}(\bar{X}_t^{i,*} - X_t^i)dW_t^i,$$

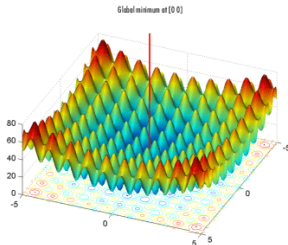
which is surely **gradient-free**.

# Numerical simulations

- A common way to set the target point  $\bar{X}_t^{i,*}$  of the  $i$ -th particle is to set the best point among the whole particles:

$$\bar{X}_t^{i,*} := \operatorname{argmin}_{x \in \{X_t^j\}} L(x), \quad \text{i.e.,} \quad L(\bar{X}_t^{i,*}) = \min_{j=1,\dots,N} L(X_t^j).$$

- The simulation ( $N = 100, \gamma = 0.01, \zeta = 0.5$ ) on **Rastrigin functions** suggests the following probabilities to find the minimum:  
In 2D and 3D, nearly 100%, however, more than 7D, less than 50%.



**Figure:** The landscape of Rastrigin function in  $d=2$  (2D).



# Table of Contents

- 1 Introduction; Consensus-Based Optimization
- 2 Random Batch Method (RBM) and its use on CBO**
- 3 Previous works; kinetic equation for CBO
- 4 Main analysis; Contraction of diameter for random network
- 5 Summary

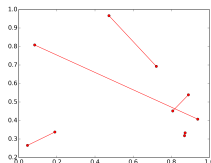
# Random Batch Method (RBM)

**Suggestion on raising success rate:** consider complex interaction structure, such as RBM (in [Carrillo–Jin–Li–Zhu 2021]).

Random Batch Method [Jin–Li–Liu, 2020] suggests a random sampling on the interactions at each time-step.

At each time  $n$ , we choose (randomly) a partition of  $\{1, \dots, N\}$  with size  $P$ , ( $2 \leq P \leq N$ ): for  $m = \lceil N/P \rceil$ ,

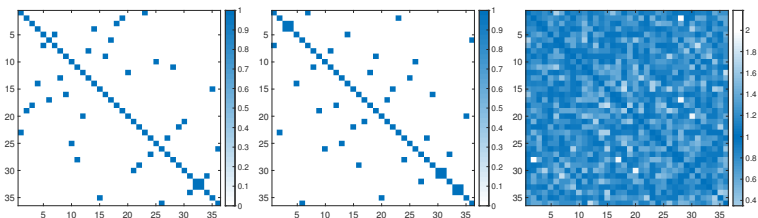
$$\{1, \dots, N\} = \mathcal{B}_1^n \cup \dots \cup \mathcal{B}_m^n, \quad |\mathcal{B}_\ell^n| = P \quad \text{for } \ell < m \quad \text{and} \quad |\mathcal{B}_m^n| \leq P.$$



**Figure:** Pairing from 10 particles ( $P = 2$ )

# Network structure of RBM

From RBM, we the interaction network becomes **sparse**:



**Figure:** Color representation of the adjacency matrix.

(Left two) Examples of random networks from RBM ( $P = 2, N = 36$ ).

(Right) An average of independent 400 matrices.

Convergence theorem [Jin–Li–Liu, 2020]

Following the law of large numbers, the solution of **RBM converges to the original (all-to-all) dynamics** as the time-discretization goes zero.

# Numerical simulations

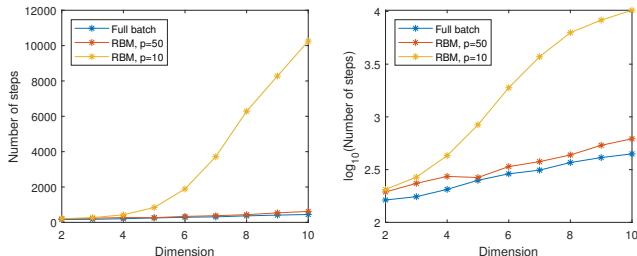
The simulation result shows that if  $P$  gets smaller,  
then the success rate grows but the cost of computation also grows.

| Success rate | Full batch ( $P = 100$ ) | $P = 50$ | $P = 10$ |
|--------------|--------------------------|----------|----------|
| $d = 2$      | 1.000                    | 1.000    | 1.000    |
| $d = 3$      | 0.988                    | 0.983    | 0.998    |
| $d = 4$      | 0.798                    | 0.920    | 0.988    |
| $d = 5$      | 0.712                    | 0.658    | 0.931    |
| $d = 6$      | 0.513                    | 0.655    | 0.880    |
| $d = 7$      | 0.388                    | 0.464    | 0.854    |
| $d = 8$      | 0.264                    | 0.389    | 0.832    |
| $d = 9$      | 0.170                    | 0.323    | 0.868    |
| $d = 10$     | 0.117                    | 0.274    | 0.886    |

Figure: Success rates from 1000 simulations.

# Numerical simulations

The accuracy of algorithm directly depends on the computation time.



**Figure:** Average number of steps until stopping criterion holds.

The stopping criterion is made with the change of positions,

$$\sum_{i=1}^N |x_{n+1}^i - x_n^i|^2 < 10^{-3}.$$

# consensus-based optimization (CBO)

Algorithm [K.–Ha–Jin–Kim 2022] based on [Carrillo–Jin–Li–Zhu 2021]

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

$$\gamma > 0, \quad \eta_t^{i,\ell} \sim \mathcal{N}(0, \sqrt{\zeta}) \quad \text{for each } i, \ell, t \quad \text{and}$$

$$\bar{X}_t^{i,*} := \operatorname{argmin}_{x \in \{X_t^j | j \in N_i(t)\}} L(x), \quad N_i(t) \text{ determined by RBM.}$$

It corresponds to a continuous SDE:

$$dX_t^i = \lambda(\bar{X}_t^{i,*} - X_t^i)dt + \sigma \text{diag}(\bar{X}_t^{i,*} - X_t^i)dW_t^i.$$

Here  $\bar{X}_t^{i,*}$  is the minimum guess from the batch of the  $i$ -th particle.

**Objectives:** here we want to guarantee the termination of the algorithm by showing convergence of the particles:  $X_t^i \rightarrow X_\infty$  as  $t \rightarrow \infty$ .

# Table of Contents

- 1 Introduction; Consensus-Based Optimization
- 2 Random Batch Method (RBM) and its use on CBO
- 3 Previous works; kinetic equation for CBO
- 4 Main analysis; Contraction of diameter for random network
- 5 Summary

# Previous works

## Proposal of algorithms & Analysis of the convergence

- [Askari-Sichani–Jalili 2013] proposed and analyzed CBO without noise
- [Pinnau–Totzeck–Tse–Martin 2017] proposed CBO with noise
- [Carrillo–Choi–Totzeck–Tse 2018] analyzed the convergence of the kinetic CBO dynamics
- [Ha–Jin–Kim 2020,2021] analyzed the convergence of a simplified CBO algorithm.
- [Fornasier–Huang–Pareschi–Sünnen 2020] proposed CBO on hypersurfaces
- [Kim–Kang–Kim–Ha–Yang 2020] proposed CBO on the Stiefel manifold
- [Carrillo–Jin–Li–Zhu 2021] proposed CBO for high dimensional problems



# Two examples in Literature

The exploration of CBO algorithm has two parts of randomness.

Algorithm with interaction network [Askari-Sichani–Jalili 2013]

$$x_i(t+1) = x_i(t) + \gamma(\bar{x}_i^*(t) - x_i(t)), \quad \bar{x}_i^*(t) = \operatorname{argmin}_{x_k(t): k \in N_i(t)} L(\cdot)$$

Algorithm for noisy trajectory [Pinnau–Totzeck–Tse–Martin 2017]

$$dX_t^i = \lambda(\bar{X}_t^* - X_t^i)dt + \sigma|\bar{X}_t^* - X_t^i|dW_t^i,$$

with

$$\bar{X}_t^* := \frac{1}{\sum_{j=1}^N e^{-\beta L(X_t^j)}} \sum_{j=1}^N e^{-\beta L(X_t^j)} X_t^j.$$

The second  $\bar{X}_t^*$  is from the **Laplace principle**, which converges to the argument minimum as  $\beta \rightarrow \infty$ . This is an **observable** in F-P dynamics.

# Algorithm in [Pinnau–Totzeck–Tse–Martin 2017]

## Algorithm [Pinnau–Totzeck–Tse–Martin 2017]

$$dX_t^i = \lambda(\bar{X}_t^* - X_t^i)dt + \sigma|\bar{X}_t^* - X_t^i|dW_t^i,$$

with

$$\bar{X}_t^* := \frac{1}{\sum_{j=1}^N e^{-\beta L(X_t^j)}} \sum_{j=1}^N e^{-\beta L(X_t^j)} X_t^j.$$

We can formally send  $N \rightarrow \infty$  to get

$$\bar{X}_t^* \rightarrow \frac{1}{\int_{\mathbb{R}^d} e^{-\beta L(x)} d\rho_t} \int_{\mathbb{R}^d} e^{-\beta L(x)} x d\rho_t, \quad \rho_t : \text{prob. measure of } X_t^i.$$

If  $L$  has a **unique minimizer  $x_*$  in the support** of  $\rho_t$ , then

$$m[\rho_t] := \frac{1}{\int_{\mathbb{R}^d} e^{-\beta L(x)} d\rho_t} \int_{\mathbb{R}^d} e^{-\beta L(x)} x d\rho_t \rightarrow x_* \quad \text{as } \beta \rightarrow \infty.$$

# Mean-field limit to the kinetic dynamics

From the mean-field limit process, the dynamics of

$$dX_t^i = \lambda(\bar{X}_t^* - X_t^i)dt + \sigma|\bar{X}_t^* - X_t^i|dW_t^i$$

becomes dynamics of the density  $\rho_t \in \mathcal{P}(\mathbb{R}^d)$  as a Fokker-Planck equation:

$$\partial_t \rho_t = \lambda \nabla \cdot ((x - m[\rho_t])\rho_t) + \frac{\sigma^2}{2} \Delta(|x - m[\rho_t]|^2 \rho_t).$$

## Theorem (Convergence) [Pinnau–Totzeck–Tse–Martin 2017]

If  $\lambda$  is large enough (compared to  $d$ ,  $\sigma^2$ , and  $e^{-\beta}$ ), then  $\mathbb{E}(\rho_t)$  converges and

$$\text{Var}(\rho_t) = O(e^{-ct}), \quad t \rightarrow \infty.$$

Idea:  $\frac{d}{dt} \text{Var}(\rho_t) = -2\lambda \text{Var}(\rho_t) + (d\sigma^2/2) \int (x - m[\rho_t])^2 d\rho_t.$

# Limitations of kinetic approach

**CBO with RBM converges to the kinetic equation** as  $\tau \rightarrow 0$  and  $N \rightarrow \infty$ . However, kinetic equation limit has **two significant limitation**.

**1 Dynamics ignores network structure:**

The particles are not-distinguishable; all-to-all interactions happen.

**2 Sample space covers the whole space:**

The whole values we need; **we already know the global minimum**.

[Fornasier–Klock–Riedl, 2021]

We should get back to the **discrete-time dynamics** in order to analyze the performance of algorithms.

# Table of Contents

- 1 Introduction; Consensus-Based Optimization
- 2 Random Batch Method (RBM) and its use on CBO
- 3 Previous works; kinetic equation for CBO
- 4 Main analysis; Contraction of diameter for random network**
- 5 Summary

# Main result: Consensus of a general CBO algorithm

For the CBO algorithm

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

we assume  $\bar{X}_t^{i,*}$  is in the convex hull of information at  $i$ :

$$\bar{X}_t^{i,*} := \sum_{j \in N_i(t)} f_{ij}(t) X_t^j : \text{convex combination}, \quad \sum_{j \in N_i(t)} f_{ij}(t) = 1, \quad \forall i.$$

## Theorem (K.-Ha-Jin-Kim 2022)

For sufficiently small (not depending on  $d$ )  $\zeta := \text{Var}(\eta_t^{i,l})$ ,  
(1) for some positive random variable  $\varepsilon$ ,

$$\mathbb{E} \max_{i,j} \|X_t^i - X_t^j\| = \mathcal{O}(e^{-\varepsilon t}), \quad t \rightarrow \infty.$$

(2) the following holds almost surely: for some positive random  $\varepsilon$ ,

$$\max_{i,j} \|X_t^i - X_t^j\| = \mathcal{O}(e^{-\varepsilon t}), \quad t \rightarrow \infty.$$

# Algorithm in [Askari-Sichani–Jalili 2013]

We may **rewrite the dynamics** without noise in [Askari-Sichani–Jalili 2013],

$$x_i(t+1) = x_i(t) + \gamma(\bar{x}_i^*(t) - x_i(t)), \quad \bar{x}_i^*(t) = \operatorname{argmin}_{x_k(t): k \in N_i(t)} L(\cdot),$$

as in the **matrix form**:

$$X(t+1) = A(t)X(t).$$

For example, if there are 4 particles and the third is the  $\bar{x}_i^*(t)$  for all  $i$ :

$$A(t) = \begin{bmatrix} 1-\gamma & 0 & \gamma & 0 \\ 0 & 1-\gamma & \gamma & 0 \\ 0 & 0 & \mathbf{1} & 0 \\ 0 & 0 & \gamma & 1-\gamma \end{bmatrix}.$$

In this case, we have

$$|x_i(t+1) - x_j(t+1)| \leq (1-\gamma)|x_i(t) - x_j(t)|.$$

# Ergodicity coefficient

**Question:** How can we prove the consensus?

**Answer:** Analysis on the **diameter** as the variance in the kinetic equation.

$$\mathcal{D}(x) := \max_{i,j} |x_i - x_j|.$$

For a stochastic matrix, define **ergodicity coefficient** as

$$\alpha(A) := \min_{i,j} \sum_k \min\{a_{ik}, a_{jk}\} \in [0, 1].$$

Note:  $\alpha(A) = 1 \Leftrightarrow$  all rows of  $A$  are identical.

**Proposition** [Askari-Sichani–Jalili 2013] from [Markov 1906]

$$\mathcal{D}(Ax) \leq (1 - \alpha(A))\mathcal{D}(x).$$



# Problem 1; the case of ergodicity zero

For example, the stochastic matrix may look like

$$A = \begin{bmatrix} 1-\gamma & \gamma & 0 & 0 \\ 0 & 1-\gamma & \gamma & 0 \\ 0 & 0 & 1-\gamma & \gamma \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \text{such as} \quad A = \begin{bmatrix} 0.9 & 0.1 & 0 & 0 \\ 0 & 0.9 & 0.1 & 0 \\ 0 & 0 & 0.9 & 0.1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Since this is a block matrix, the ergodicity constant should be zero:

$$\alpha(A) := \min_{i,j} \sum_k \min\{a_{ik}, a_{jk}\} = 0. \quad (i=1, j=3)$$

However, the network is connected; if we consider  $A^2$  and  $A^3$ ,

$$A^2 = \begin{bmatrix} 0.81 & 0.18 & 0.01 & 0 \\ 0 & 0.81 & 0.18 & 0.01 \\ 0 & 0 & 0.81 & 0.19 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A^3 = \begin{bmatrix} 0.729 & 0.243 & 0.027 & 0.001 \\ 0 & 0.729 & 0.243 & 0.028 \\ 0 & 0 & 0.729 & 0.271 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Therefore,  $\alpha(A^3) \neq 0$  since  $i=1$  and  $j=4$  has length 3 connectivity.

# Analytical result in [Askari-Sichani–Jalili 2013]

Therefore, if the network is **connect** so that

$$\lim_{n \rightarrow \infty} \alpha(A^n) = 1,$$

then  $\mathcal{D}(A^n x) \leq (1 - \alpha(A^n))\mathcal{D}(x) \rightarrow 0$ .

It is convenient to check the convergence to 1 as follows. Define

$$A((t, s]) := A(t-1)A(t-2) \dots A(s) \quad (t > s).$$

## Proposition [Askari-Sichani–Jalili 2013]

Assume that there exists  $0 = t_0 < t_1 < t_2 < \dots$  satisfying

$$\sum_{i=1}^{\infty} \alpha(A((t_i, t_{i-1}])) = \infty.$$

Then  $\lim_{n \rightarrow \infty} \alpha(A^n) = 1$  and  $\mathcal{D}(X(t)) \rightarrow 0$  as  $t \rightarrow \infty$ .

## Problem 2; connectivity of RBM

The problem occurs when  $\bar{x}_i^* \neq \bar{x}_j^*$ .

For example, the stochastic matrix may look like

$$A(t) = \begin{bmatrix} 1-\gamma & \gamma & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1-\gamma & \gamma \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Since this is a block matrix, the ergodicity constant should be zero:

$$\alpha(A) := \min_{i,j} \sum_k \min\{a_{ik}, a_{jk}\} = 0. \quad (i=1, j=3)$$

Therefore, the random network needs connectivity.

**Lemma: positive ergodicity of a network**

Assume that for any  $i$  and  $j$ , suppose that **the information of  $i$  at time  $t$  is observed by  $j$  at time  $t+m$** . Then,  $\alpha(A((s+m, s])) \geq \gamma(1-\gamma)^m$ .

# Consensus with RBM

Now we may prove that the diameter decays almost surely.

First, from **ergodicity** argument, we have

$$\mathcal{D}(X(t_{km})) \leq \exp\left(-\gamma(1-\gamma)^m \sum_{s=1}^k \mathcal{G}_{s,m}\right) \mathcal{D}(X(0)),$$

where  $\mathcal{G}_{s,m}$  is 1 if all particles are connected by length  $m$ , and 0 otherwise. (This is a **random variable** with positive probability to be 1)

Note that

$$\lim_{k \rightarrow \infty} \frac{1}{k} \sum_{s=1}^k \mathcal{G}_{s,m} = \mathbb{E}[\mathcal{G}_{s,m}] = p_m > 0.$$

Therefore, we conclude the decay of the diameter

$$\begin{aligned} \mathcal{D}(X(t_{km})) &\leq \exp(-\Lambda(m, k)k) \mathcal{D}(X(0)), \\ \lim_{k \rightarrow \infty} \Lambda(m, k) &= \gamma(1-\gamma)^m p_m. \end{aligned}$$

In other words, **the algorithm satisfy consensus**.

# Ergodicity with noise?

Now we consider noise;

$$X_{(t+1)}^i = X_t^i + \gamma(\bar{X}_t^{i,*} - X_t^i) + \text{diag}(\eta_t^{i,1}, \dots, \eta_t^{i,d})(\bar{X}_t^{i,*} - X_t^i),$$

Then, the network structure

$$A(t) = \begin{bmatrix} 1 - \gamma & \gamma & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 - \gamma & \gamma \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

becomes

$$A_\varepsilon(t) = \begin{bmatrix} 1 - \gamma + \varepsilon & \gamma - \varepsilon & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 - \gamma + \varepsilon & \gamma - \varepsilon \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Then,  $A$  is **stochastic** but can have **negative ergodicity**.

# Table of Contents

- 1 Introduction; Consensus-Based Optimization
- 2 Random Batch Method (RBM) and its use on CBO
- 3 Previous works; kinetic equation for CBO
- 4 Main analysis; Contraction of diameter for random network
- 5 Summary**

# Summary and remaining questions

## Remarks

- 1** The convergence of CBO holds with random batch method.
- 2** This is the first result on the convergence of CBO with individual random motion on particles.
- 3** No performance result yet on the global minimizer.

Thank you very much